Concavity, Curve Sketching, and Optimization

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Increasing/decreasing and the first derivative test

For $f(x) = x^3 - 6x^2 + 9x$, find the intervals where f is increasing and decreasing, and classify each critical point using the First Derivative Test.

2 [Foundational] • Target: Concavity and the second derivative test

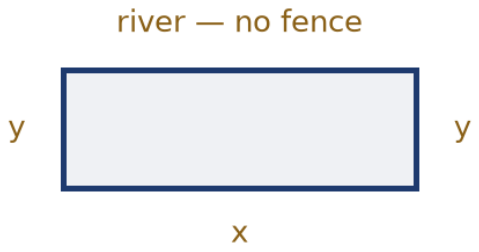
For $f(x) = x^4 - 4x^3$, find the intervals of concavity and any inflection points.

3 [Foundational] • Target: Curve sketching

For $f(x) = x^3 - 3x^2$, find: (a) critical points and local extrema, (b) inflection points, (c) intervals of increase/decrease and concavity. Summarize with a sign chart.

4 [Foundational] • Target: Optimization problems

A farmer has 200 m of fencing to enclose a rectangular field bordering a river (no fence needed along the river). Find the dimensions that maximize the enclosed area.



5 [Foundational] • Target: Optimization problems

An open-top box is made from a 12 in by 12 in square sheet by cutting equal squares of side x from each corner and folding up the sides. Find x that maximizes the volume.

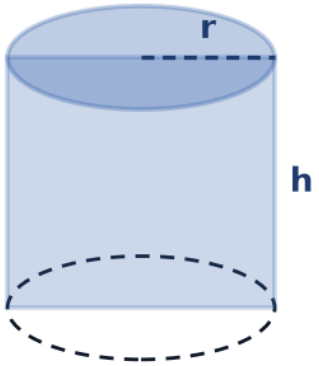
6 [Foundational] • Target: Optimization: minimizing cost or distance

Find the point on the line $y = 2x - 3$ closest to the origin, by minimizing the squared distance $D = x^2 + y^2$.

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7 [Foundational] • Target: Optimization: minimizing cost or distance

A closed cylindrical can must have volume 16π cm^3. Using $V = \pi r^2 h$ and surface area $S = 2\pi r^2 + 2\pi rh$, find the radius that minimizes surface area.



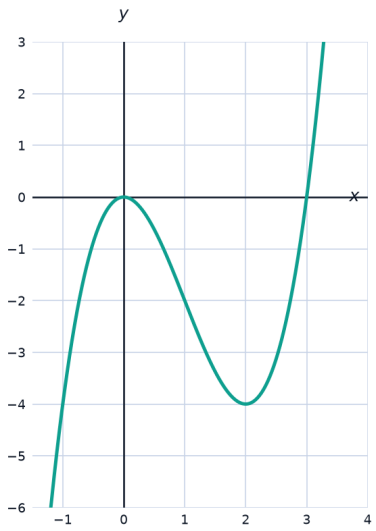
Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Points of inflection and graph behavior

Explain why a point of inflection does NOT necessarily correspond to a local maximum or minimum, using $f(x) = x^3$ at $x = 0$ as an example.

9 [Intermediate] • Target: Visualizing extrema and concavity together

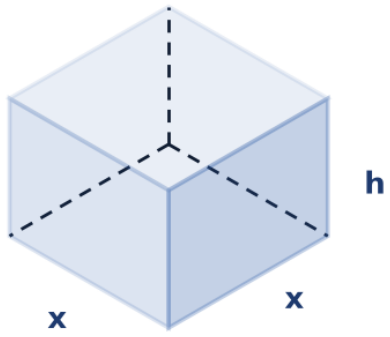
The graph shows $f(x) = x^3 - 3x^2$ from the curve-sketching problem above, with its local max at $x = 0$, local min at $x = 2$, and inflection point at $x = 1$.



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10 [Intermediate] • Target: A capstone optimization problem

A rectangular box with a square base and no top must have volume 500 cm^3. Using $V = x^2h = 500$ (so $h = \frac{500}{x^2}$) and surface area $S = x^2 + 4xh$, find the base side length x that minimizes the surface area.



11 [Intermediate] • Target: Second derivative test failures

For $f(x) = x^4$, show that the Second Derivative Test is inconclusive at the critical point $x = 0$, then use the First Derivative Test to correctly classify it.

12 [Intermediate] • Target: A capstone curve-sketching problem

For $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$, find all critical points and classify each using the Second Derivative Test.

13 [Intermediate] • Target: Curve sketching

Level 2 · Medium Solve the initial value problem $\frac{dy}{dx} - y = e^{3x}$ with $y(0) = 2$.

14 [Intermediate] • Target: Optimization problems

Level 2 · MediumApply the Ratio Test to determine the convergence or divergence of $\sum_{n=1}^{\infty} \frac{4^n}{n!}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Optimization: minimizing cost or distance • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Level 1 · Foundations Evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = abla(x^3y + 2z)$ along any curve C from $(0, 0, 1)$ to $(2, 1, 3)$.

16 [Challenging] • Target: Points of inflection and graph behavior • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Level 1 · Foundations Use the Divergence Test to determine whether $\sum_{n=1}^{\infty} \cos\left(\frac{1}{n}\right)$ converges or diverges.

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17 [Challenging] • Target: Visualizing extrema and concavity together • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Level 2 · MediumFind the angle(s) $\theta \in [0, \pi]$ at which the polar curve $r = 3\cos\theta$ passes through the origin, and determine the equation of the tangent line at that point.

18 [Challenging] • Target: A capstone optimization problem • Cross-Lesson: 8.01 Area Between Curves

Find the perpendicular distance from point $P(3,\ 1,\ -2)$ to plane $\pi : 2x - 2y + z = 5$.

19 [Challenging] • Target: Second derivative test failures • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Level 1 · Foundations Find $\frac{dy}{dx}$ in terms of t for the parametric curve $x = 2\cos t$, $y = 5\sin t$ for $t \in (0, \pi)$.

20 [Challenging] • Target: A capstone curve-sketching problem • Cross-Lesson: 1.01 Limits Graphically and Numerically

Level 3 · Advanced Find the exact surface area generated by revolving the line segment $y = 2x$ on $[0, 3]$ about the x-axis.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Increasing/decreasing and the first derivative test

Classify the critical point $x = -1$ of $f(x) = x^3 + 6x^2 + 9x$, using the Second Derivative Test.

a. a local maximum

b. cannot be determined without more information

c. a local minimum

d. an inflection point

22 [Foundational] • Target: Concavity and the second derivative test

Find the nonzero inflection point of $f(x) = x^4 + 6x^3$.

a. -3

b. -6

c. -2

d. 6

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23 [Foundational] • Target: Curve sketching

A farmer has 100 m of fencing to enclose a rectangular field with $x + 2y = 100$ (one side of length x needs no fence, the other 2 side(s) of length y each do). Find y that maximizes the enclosed area $A = xy$.

a. 26

b. 100

c. 25

d. 50

24 [Foundational] • Target: Optimization problems

Find the x-coordinate of the point on the line $y = -5x - 4$ closest to the origin, by minimizing $D = x^2 + y^2$.

a. $-\frac{4}{5}$

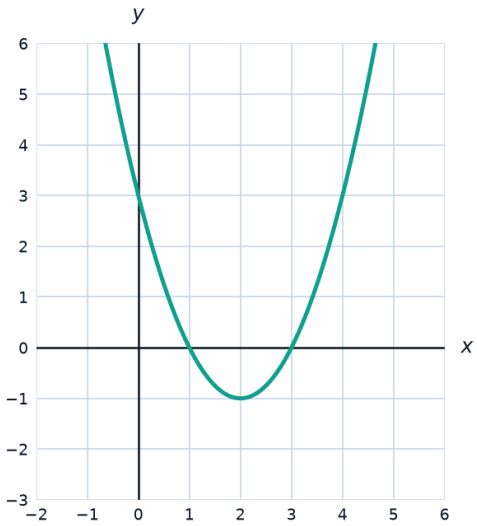
b. $\frac{2}{13}$

c. $-\frac{10}{13}$

d. $\frac{10}{13}$

25 [Foundational] • Target: Optimization: minimizing cost or distance

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is shown below. The graph intersects the x-axis at points A and B. Determine the distance between A and B.



a. 2

b. 3

c. 4

d. 5

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26 [Foundational] • Target: Points of inflection and graph behavior

Local min at $x = c$ by First Derivative Test requires f':

a. positive then negative

b. negative then positive

c. positive everywhere

d. zero everywhere

27 [Foundational] • Target: Visualizing extrema and concavity together

$f(x, y) = xy$: at origin, the second derivative test gives $D = f_{xx}f_{yy} - f_{xy}^2 = 0 - 1 = -1 < 0 \Rightarrow$

a. saddle point

b. local max

c. local min

d. inconclusive

28 [Foundational] • Target: A capstone optimization problem

If $f'(x) = 0$ and $f''(x) = 0$ at $x = c$, the Second Derivative Test is:

a. guaranteed a maximum

b. guaranteed an inflection point

c. guaranteed a minimum

d. inconclusive

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Second derivative test failures

The First Derivative Test classifies a critical point c as a local max when f' changes from:

a. positive to positive

b. negative to positive

c. zero to zero

d. positive to negative

30 [Intermediate] • Target: A capstone curve-sketching problem

The Second Derivative Test says $f''(c) < 0$ at a critical point c implies:

a. inflection point

b. no conclusion

c. local maximum

d. local minimum

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31 [Intermediate] • Target: Increasing/decreasing and the first derivative test

If $f'(x) = 0$ and $f''(x) = 0$, the Second Derivative Test is:

a. inconclusive

b. proves local min

c. proves local max

d. proves inflection

32 [Intermediate] • Target: Concavity and the second derivative test

Second Derivative Test: $f'(c) = 0$ and $f''(c) > 0 \Rightarrow x = c$ is a:

a. local max

b. local min

c. inflection point

d. saddle point

33 [Intermediate] • Target: Curve sketching

The second derivative test fails ($D = 0$). This means the critical point is:

a. always a saddle

b. indeterminate without further analysis

c. always min

d. always max

34 [Intermediate] • Target: Optimization problems

Second derivative test: $f''(c) > 0$ implies $x = c$ is:

a. local min

b. local max

c. inflection

d. saddle

35 [Intermediate] • Target: Optimization: minimizing cost or distance

The second derivative test: $D = f_{xx}f_{yy} - f_{xy}^2$. If $D > 0$ and $f_{xx} > 0$, the point is a:

a. saddle point

b. local min

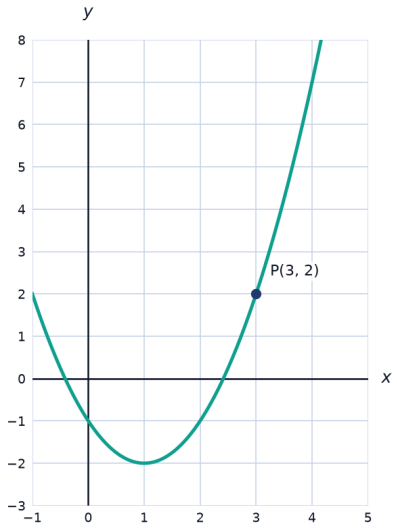
c. local max

d. inflection point

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36 [Intermediate] • Target: Points of inflection and graph behavior

The figure shows the graph of the function $f(x) = x^2 - 2x - 1$ and the point $P(3, 2)$ on the curve. What is the slope of the tangent line to the graph of f at point P?



a. 4

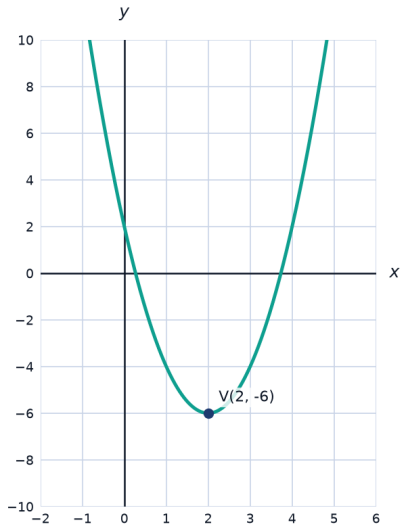
b. 2

c. 3

d. 6

37 [Intermediate] • Target: Visualizing extrema and concavity together

The diagram shows the graph of the function $f(x) = 2x^2 - 8x + k$. The graph intersects the x-axis at points A and B, and the vertex of the parabola is at point V. Given that the x-coordinate of the vertex is 2, and the y-coordinate of the vertex is -6, find the value of k.



a. 1

b. -2

c. 2

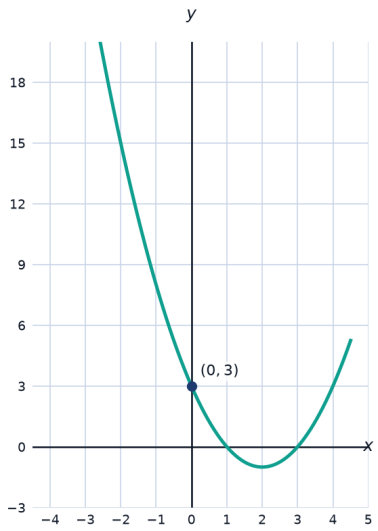
d. -1

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: A capstone optimization problem** • **Cross-Lesson: 8.01 Area Between Curves**

The graph of the quadratic function $f(x) = x^2 - 4x + 3$ is shown below. The graph intersects the x-axis at points A and B. Determine the x-coordinate of the vertex of the parabola.



a. 1

b. 2

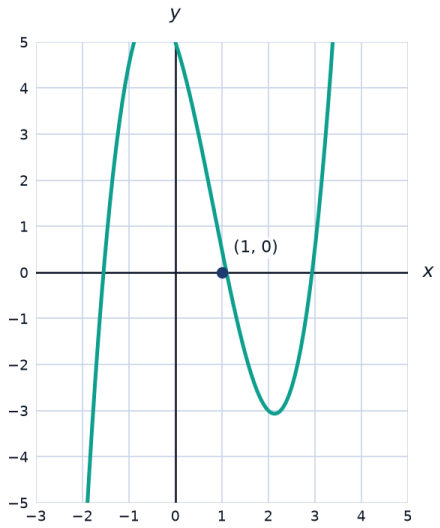
c. 3

d. 4

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39 [Challenging] • Target: Second derivative test failures • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Consider the graph of the function $f(x)$ shown below. Which of the following points lies on the graph of $f(x)$?



a. $(-2, 3)$

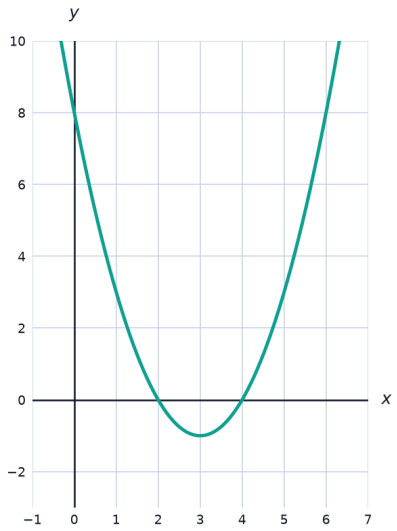
b. $(0, -1)$

c. $(1, 0)$

d. $(3, 4)$

40 [Challenging] • Target: A capstone curve-sketching problem • Cross-Lesson: 1.01 Limits Graphically and Numerically

Consider the function $f(x) = x^2 - 6x + 8$. The graph of $y = f(x)$ intersects the x-axis at points A and B, and the y-axis at point C. Find the length of the line segment AB.



a. 2 units

b. 3 units

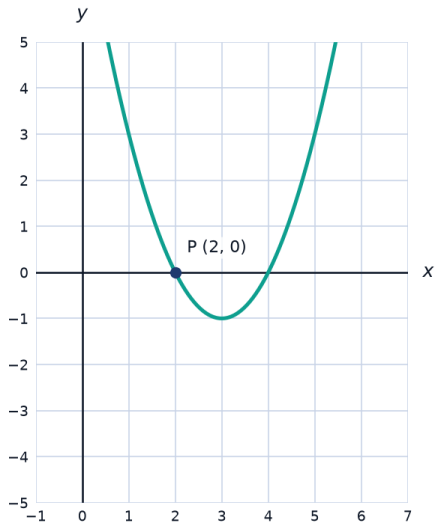
c. 4 units

d. 6 units

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41 **[Challenging]** • **Target: Increasing/decreasing and the first derivative test** • **Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem**

The graph of the function $g(x) = x^2 - 6x + k$ is shown below. The graph intersects the x-axis at points P and Q. Point P has coordinates $(2, 0)$. Find the value of the parameter k.



a. 1

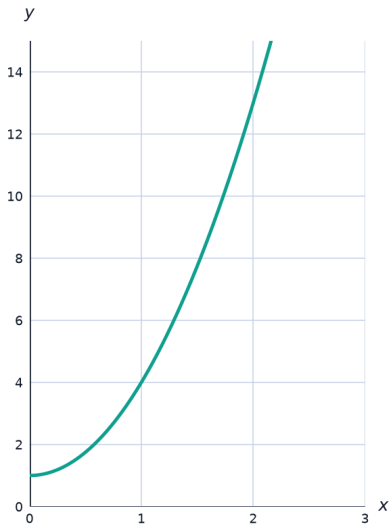
b. 2

c. 4

d. 8

42 **[Challenging]** • **Target: Concavity and the second derivative test** • **Cross-Lesson: 2.01 The Definition of the Derivative**

A region in the first quadrant is bounded by the graph of $f(x) = 3x^2 + 1$, the x-axis, and the vertical lines $x = 1$ and $x = 2$. What is the area of this region?



a. 10

b. 12

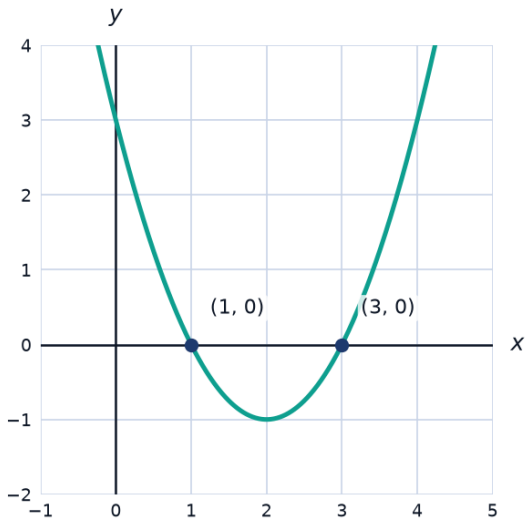
c. 14

d. 16

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43 [Challenging] • Target: Curve sketching • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

The graph of the quadratic function $f(x) = x^2 - 4x + 3$ is shown below. The graph intersects the x-axis at points where $f(x) = 0$. Determine the sum of the x-coordinates of these intersection points.



a. 1

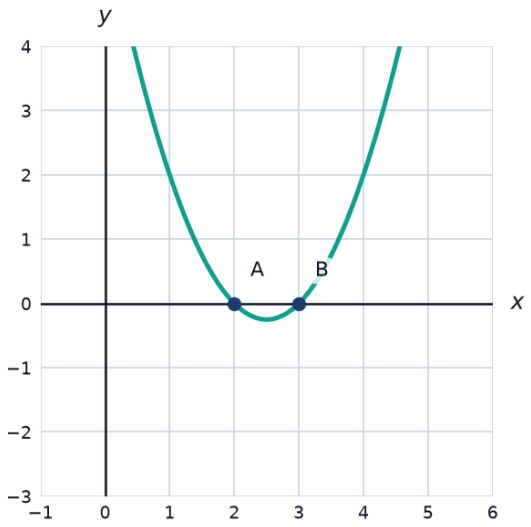
b. 3

c. 4

d. -3

44 [Challenging] • Target: Optimization problems • Cross-Lesson: 3.01 The Chain Rule

The graph of the quadratic function $f(x) = x^2 - 5x + 6$ is shown below. The graph intersects the x-axis at points A and B. Find the sum of the x-coordinates of points A and B.



a. 5

b. -5

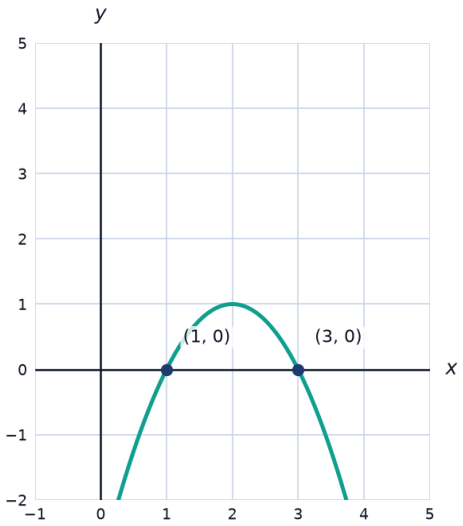
c. 6

d. -6

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45 [Challenging] • Target: Optimization: minimizing cost or distance • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

Consider the function $f(x) = -x^2 + 4x - 3$. The graph of $y = f(x)$ is shown below. The graph intersects the x-axis at two points. What is the value of the x-coordinate of the point of intersection with the larger x-value?



a. 1

b. 2

c. 3

d. 4

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